

CS & IT ENGINEERING



Digital Logic

Combinational Circuits

Lecture No. 06



By- Aditya sir

Recap of Previous Lecture



Topic



Set 2 ✓
Set 3 ✓ → 90%
Set 4

Topics to be Covered



Topic

Complete MUX





About Aditya Jain sir

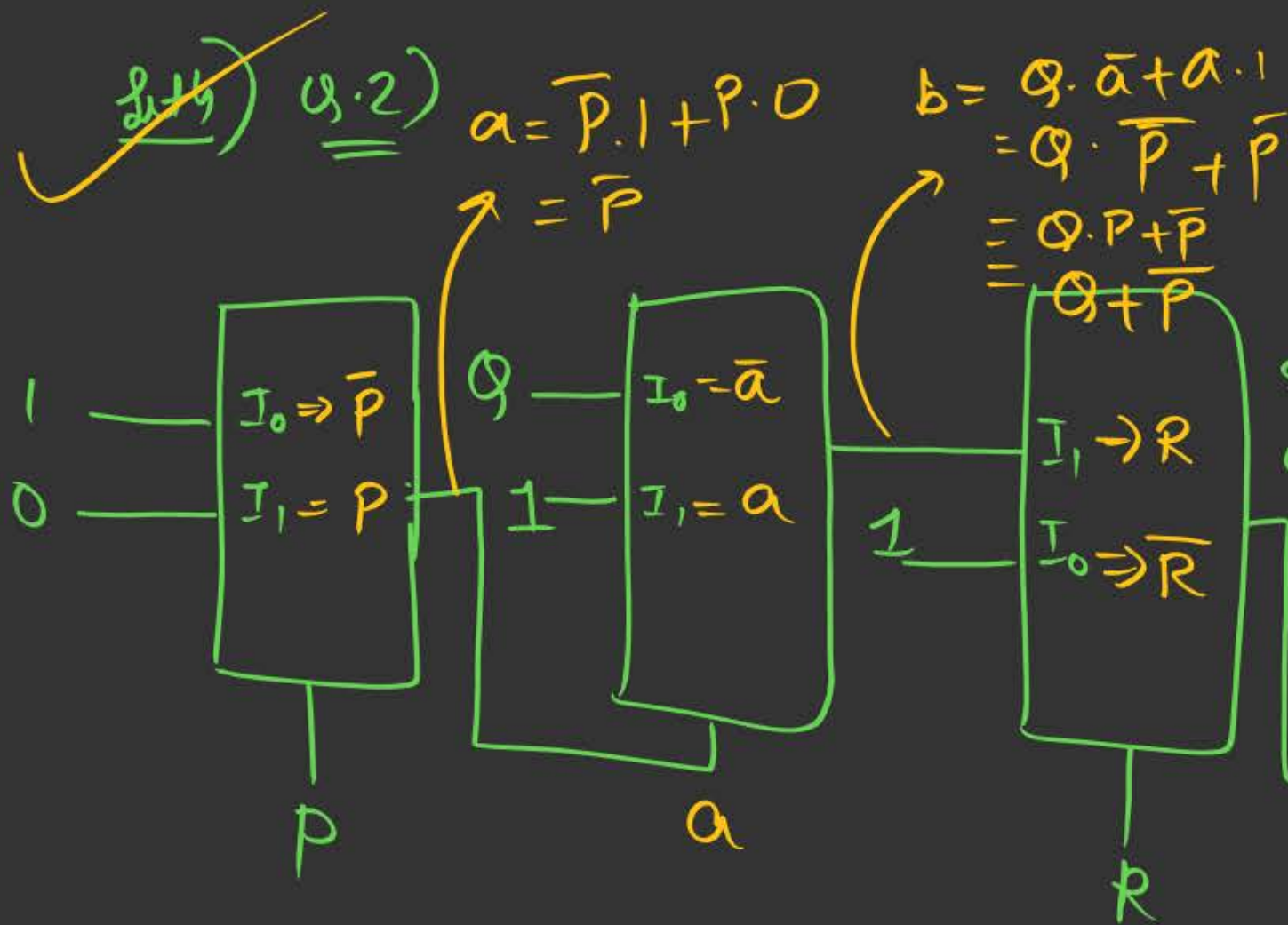


1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.





Telegram Link for Aditya Jain sir: [https://t.me/AdityaSir PW](https://t.me/AdityaSir_PW)



min NAND

$X = ?$

$$X = S \cdot \bar{C} + 0 \cdot C = S \cdot \bar{C}$$

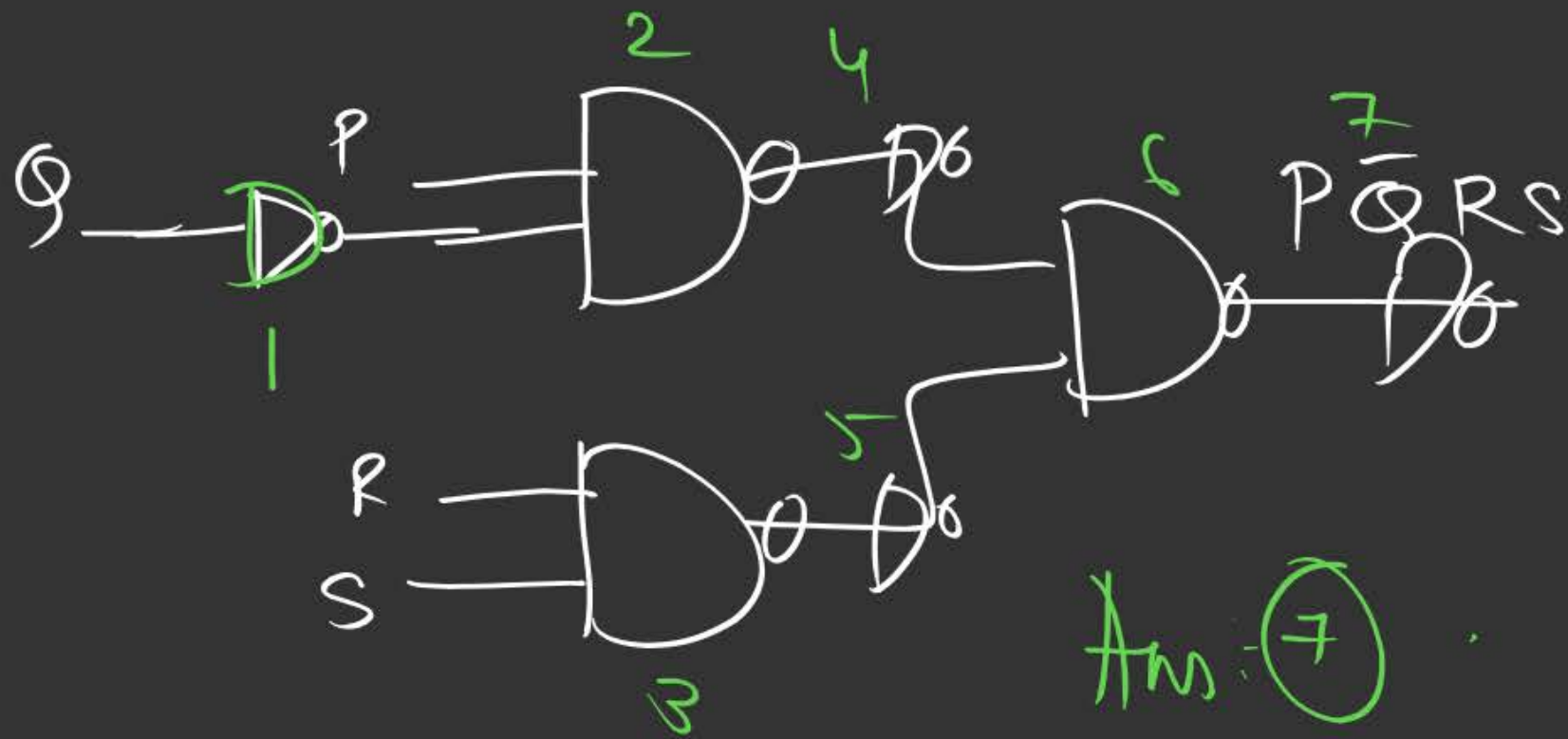
$$\begin{aligned} C &= b \cdot R + 1 \cdot \bar{R} \\ &= b + \bar{R} \\ &= (Q + \bar{P} + \bar{R}) \end{aligned}$$

$$\begin{aligned} &= S \cdot (Q + \bar{P} + \bar{R}) \\ &= S \cdot (\bar{Q} \cdot \bar{P} \cdot \bar{R}) \\ &= (P \cdot \bar{Q} \cdot R \cdot S) \end{aligned}$$

Part-2 :- min no. of NAND:

$$n=4, k=1$$

Ans:- $X = P \bar{Q} R S$



Ans: 7

NAND

$$(2n-2)+k$$

$$(4 \times 2 - 2) + 1$$

$$(8-2)+1$$

$$= 6+1$$

$$= \underline{7}$$

NOR

$$(3n-3)-k$$

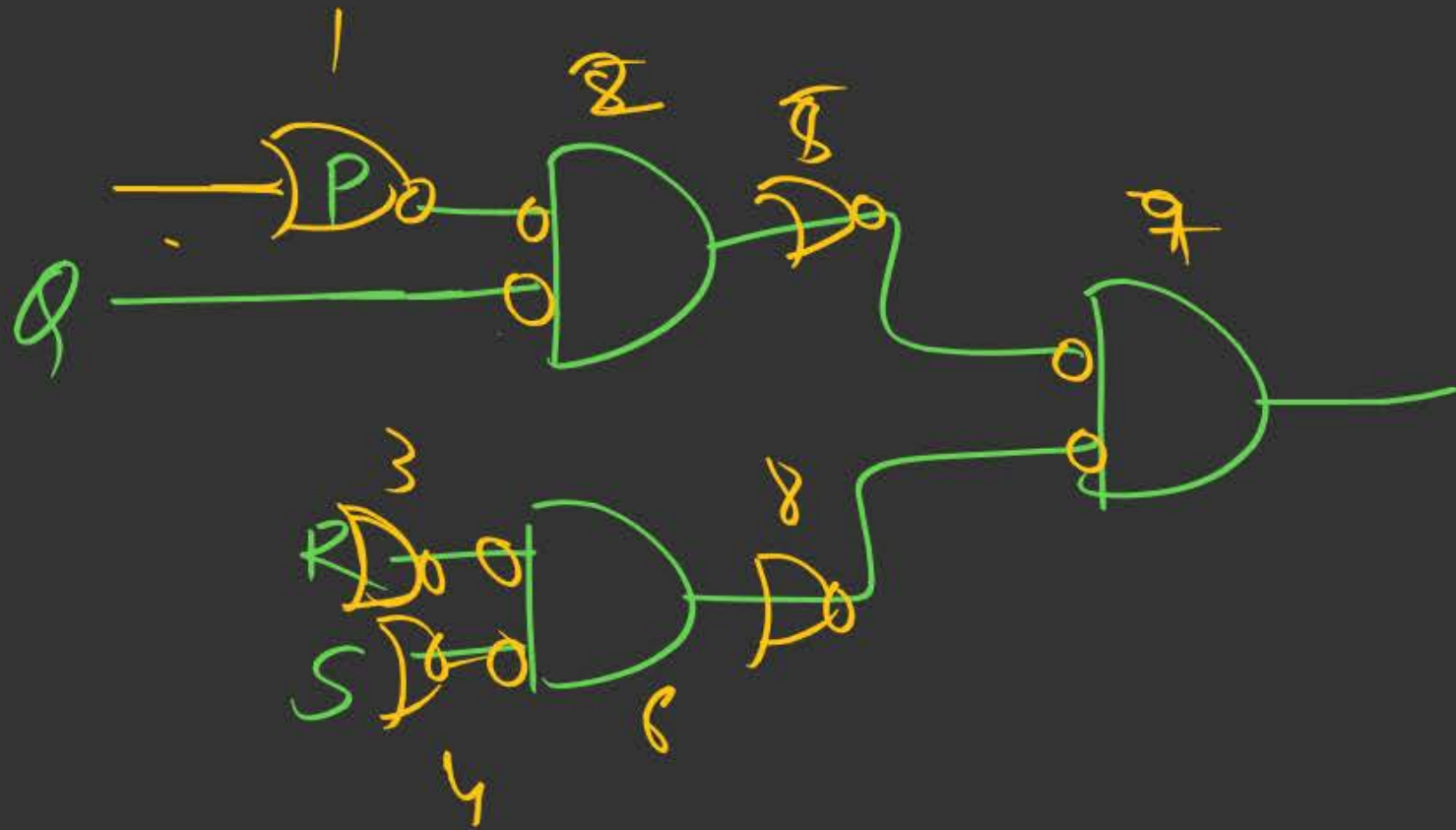
$$= 3 \times 4 - 3 - 1$$

$$= 12 - 3 - 1$$

$$= \underline{8} \checkmark$$

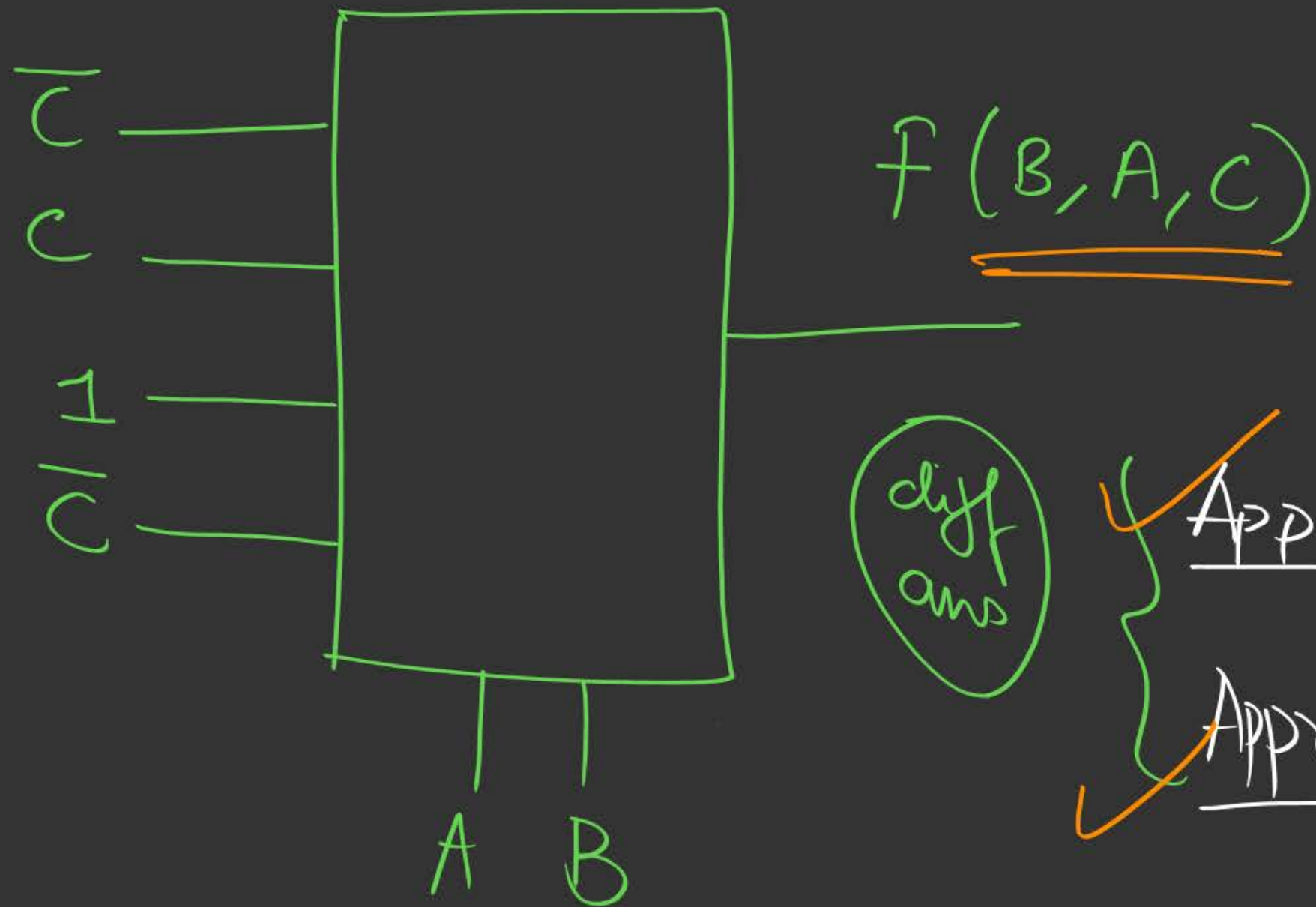
Part-3: min no. of NOR gates?

$$x = P\bar{Q}RS$$



Total 8 NOR gates

Sef 3: (Variable ordering)



$\Rightarrow$ 2 ways /
interpretation

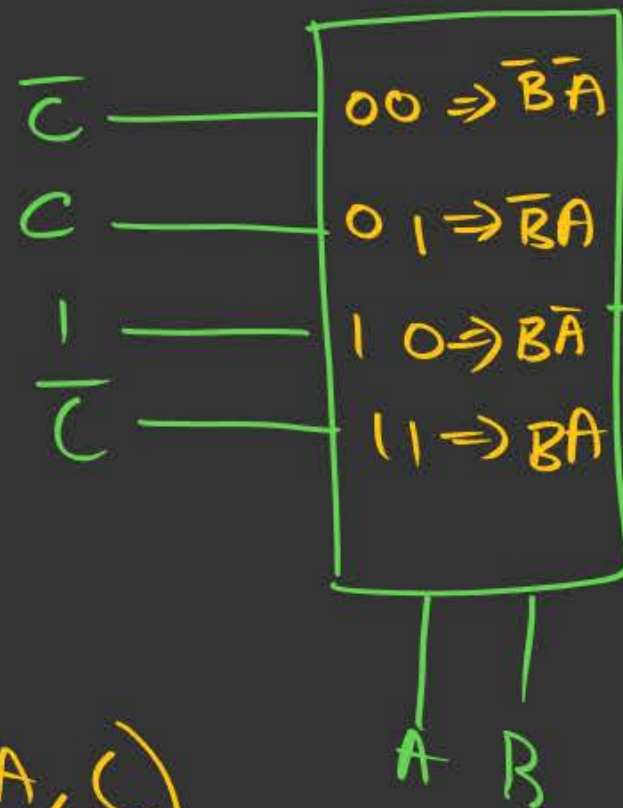
App 01 $\rightarrow$ default case

App 02 $\rightarrow$ Specific select line ordering mentioned

Appr 1: Default (no requirement specified for select line ordering)

Final ordering for SCF (SOP)

(both apprs)



$F(B, A, C)$

$$= \bar{B}\bar{A} \cdot \bar{C} + \bar{B}\bar{A} \cdot C + B\bar{A} \cdot 1 + BA \cdot \bar{C}$$

$$= \bar{B}\bar{A}\bar{C} + \bar{B}\bar{A}C + B\bar{A}(C + \bar{C}) + BA\bar{C}$$

$$= \bar{B}\bar{A}\bar{C} + \bar{B}\bar{A}C + B\bar{A}\bar{C} + B\bar{A}C + BA\bar{C}$$

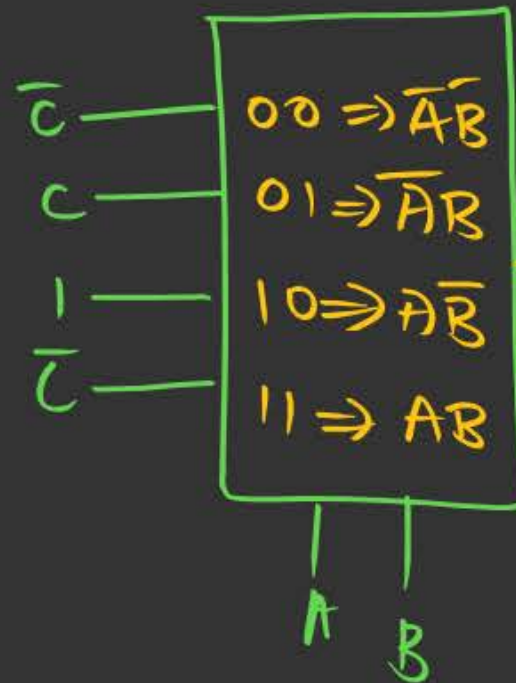
$$000 \quad 011 \quad 100 \quad 101 \quad 110$$

$$= \Sigma(0, 3, 4, 5, 6) \checkmark$$

$F(B, A, C)$

$\rightarrow$ then BAC

Appr 2 :- When select line ordering is specifically mentioned to be followed.



$F(B, A, C)$

$$= \bar{A}\bar{B} \cdot \bar{C} + \bar{A}B\bar{C} + A\bar{B} \cdot 1 + AB\bar{C}$$

$$= \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}(1 + \bar{C}) + AB\bar{C}$$

$$= \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C}$$

$$\Rightarrow \bar{B}\bar{A}\bar{C} + B\bar{A}\bar{C} + \bar{B}A\bar{C} + \bar{B}A\bar{C} + BA\bar{C}$$

$$000 \quad 101 \quad 010 \quad 011 \quad 110$$

$$= \Sigma(0, 2, 3, 5, 6)$$

Ordering

$f(B, A, C)$

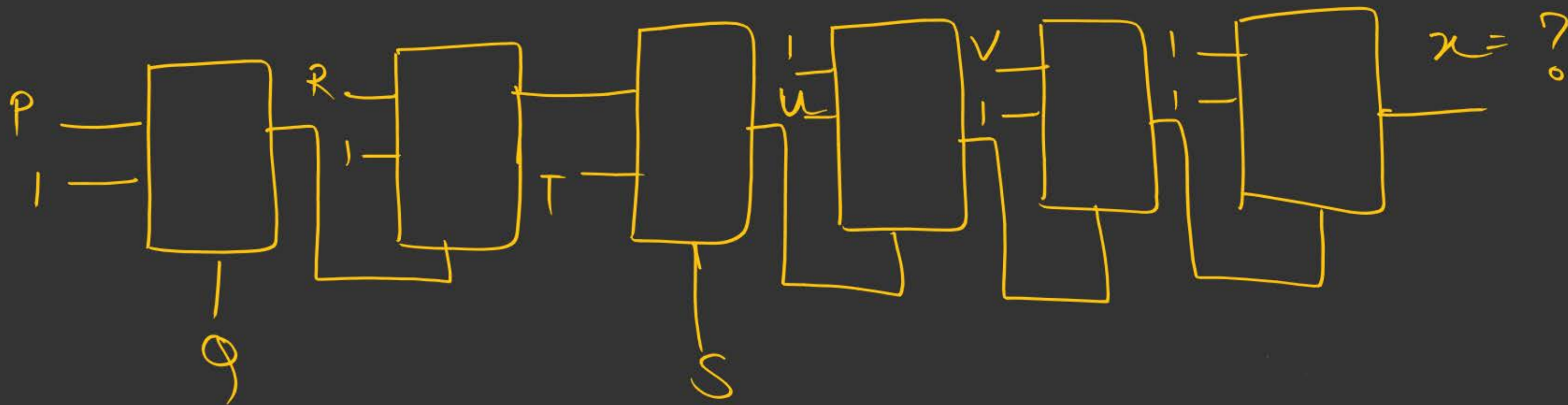
(prover eg)

default

$x_3 = 11 = AB$	
$x_0 = 00 = \bar{A}\bar{B}$	
$x_2 = 10 = A\bar{B}$	
$x_1 = 01 = \bar{A}B$	
	A B

00	$\bar{A}\bar{B}$
01	$\bar{A}B$
10	$A\bar{B}$
11	AB
	A B

$$x = f(P, Q, R, S, T, U, V)$$



$$x = f(P, Q, R, S, T, U, V)$$

Appr 1:-

$$b = R\bar{a} + a$$

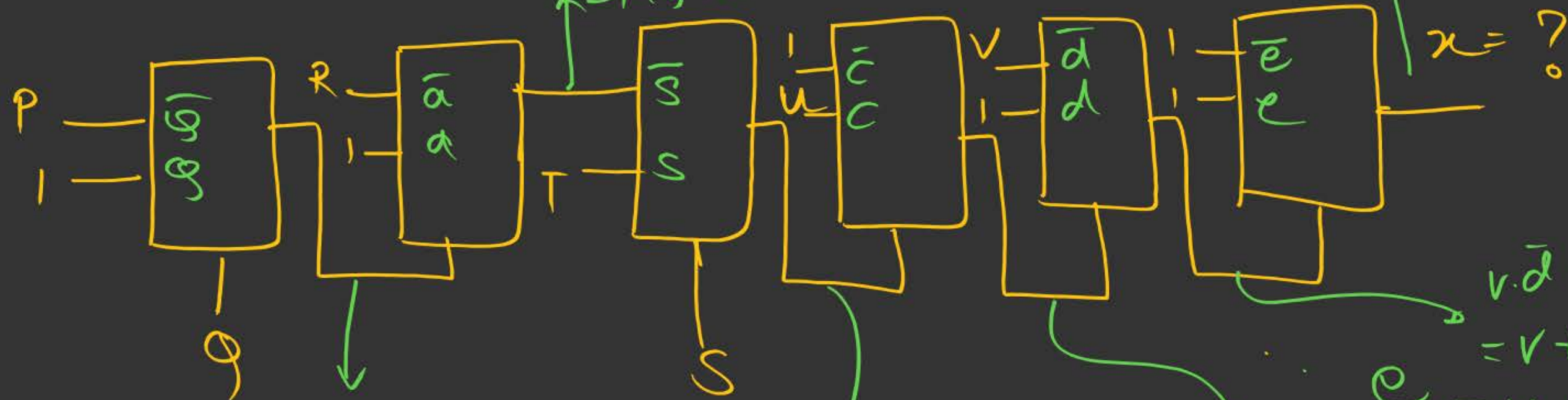
$$= R + a$$

$$= P + Q + R$$

$$x = \bar{e} \cdot 1 + e \cdot 1$$

$$= \bar{e} + e = \textcircled{1}$$

010
0



$$a = P\bar{Q} + Q$$

$$= P + Q$$

$$c = b\bar{S} + T \cdot S$$

$$= (P + Q + R)\bar{S} + T \cdot S$$

$$d = \bar{c} + u \cdot c$$

$$= \bar{c} + u$$

$$= \overline{(P + Q + R)\bar{S} + T \cdot S} + u$$

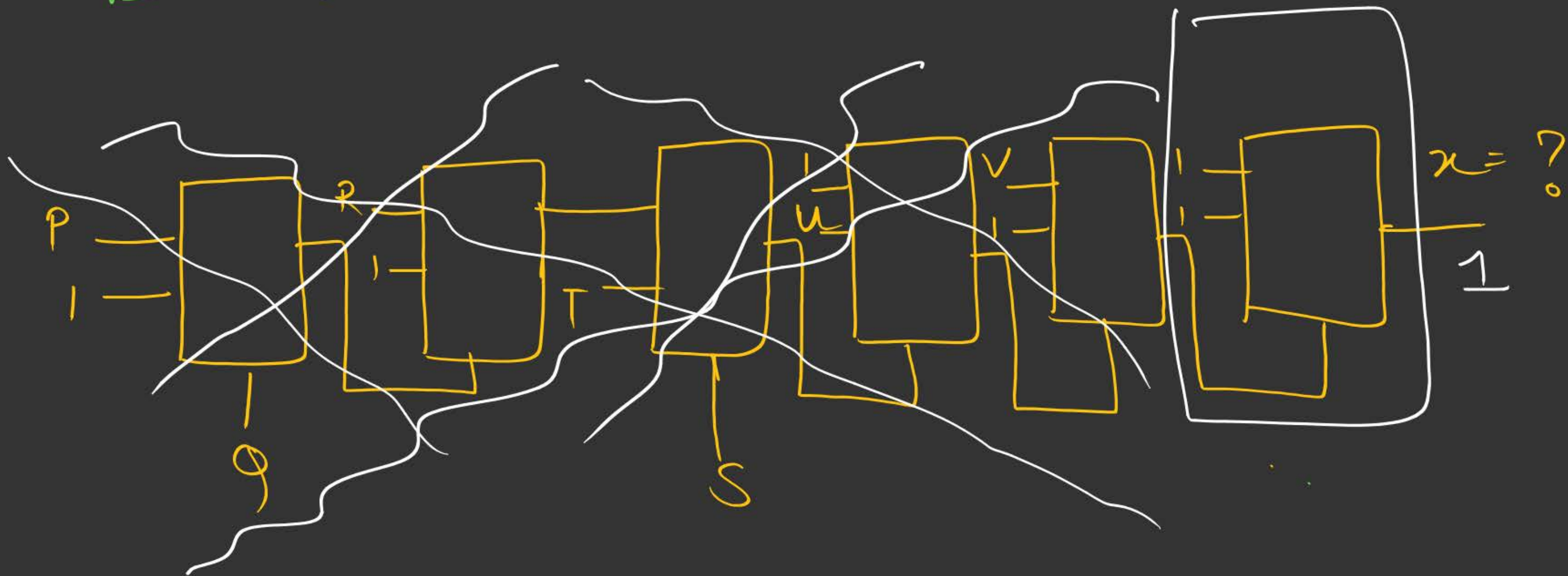
$$v \cdot \bar{d} + d$$

$$= v + d$$

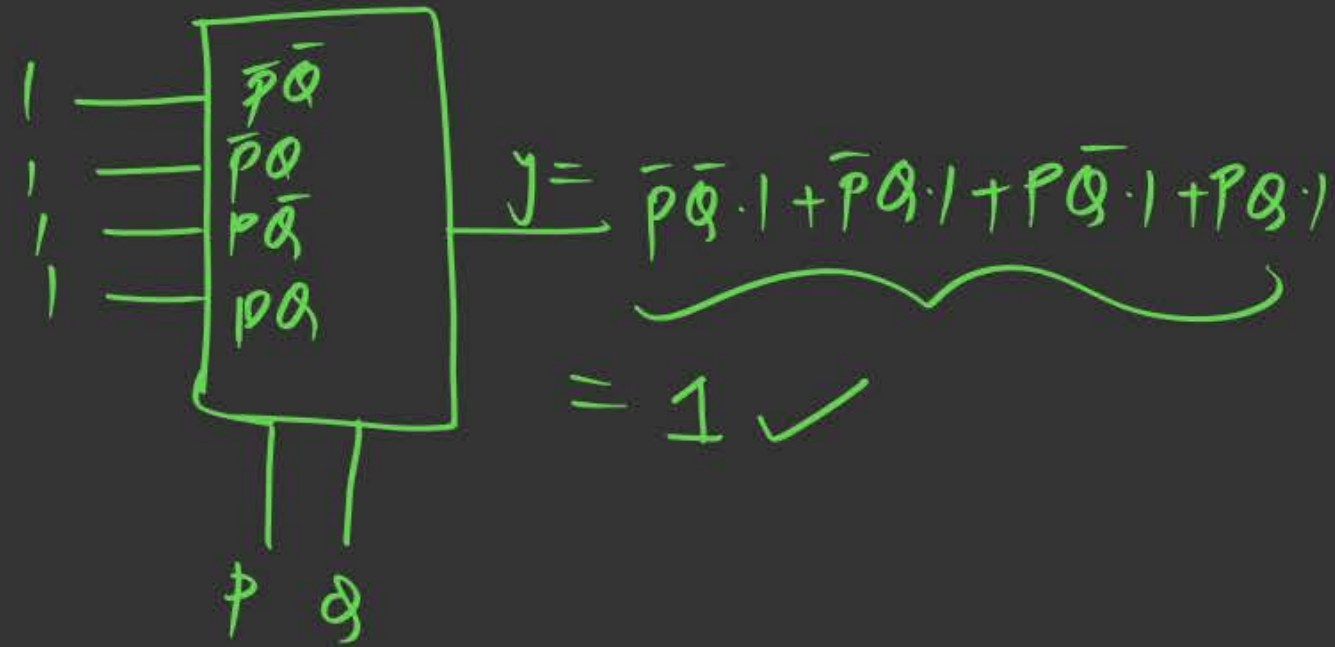
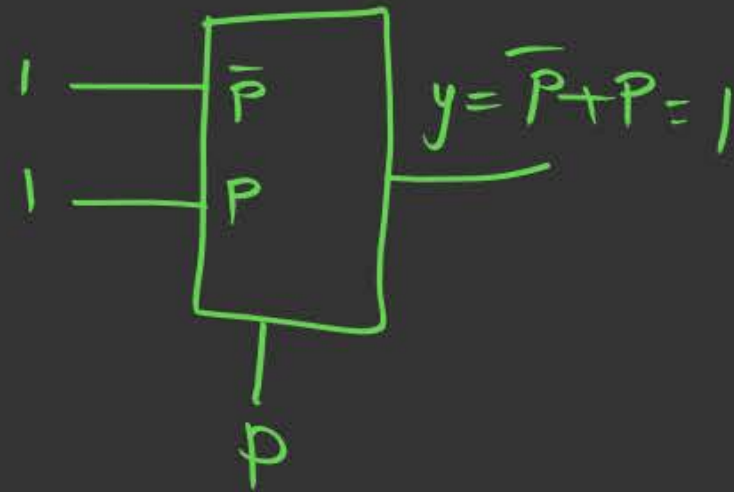
$$e = V + (\bar{P} \bar{Q} \bar{R}) + S \cdot (\bar{T} + \bar{S}) + u$$

$x = ?$

Appx 2:- $x = f(P, Q, R, S, T, U, V)$



Appr 2 Concept :-



Note :- When all i/p of MUX = 1, then o/p is 1

Set 3 ✓

Set 4 ✓

Set 5

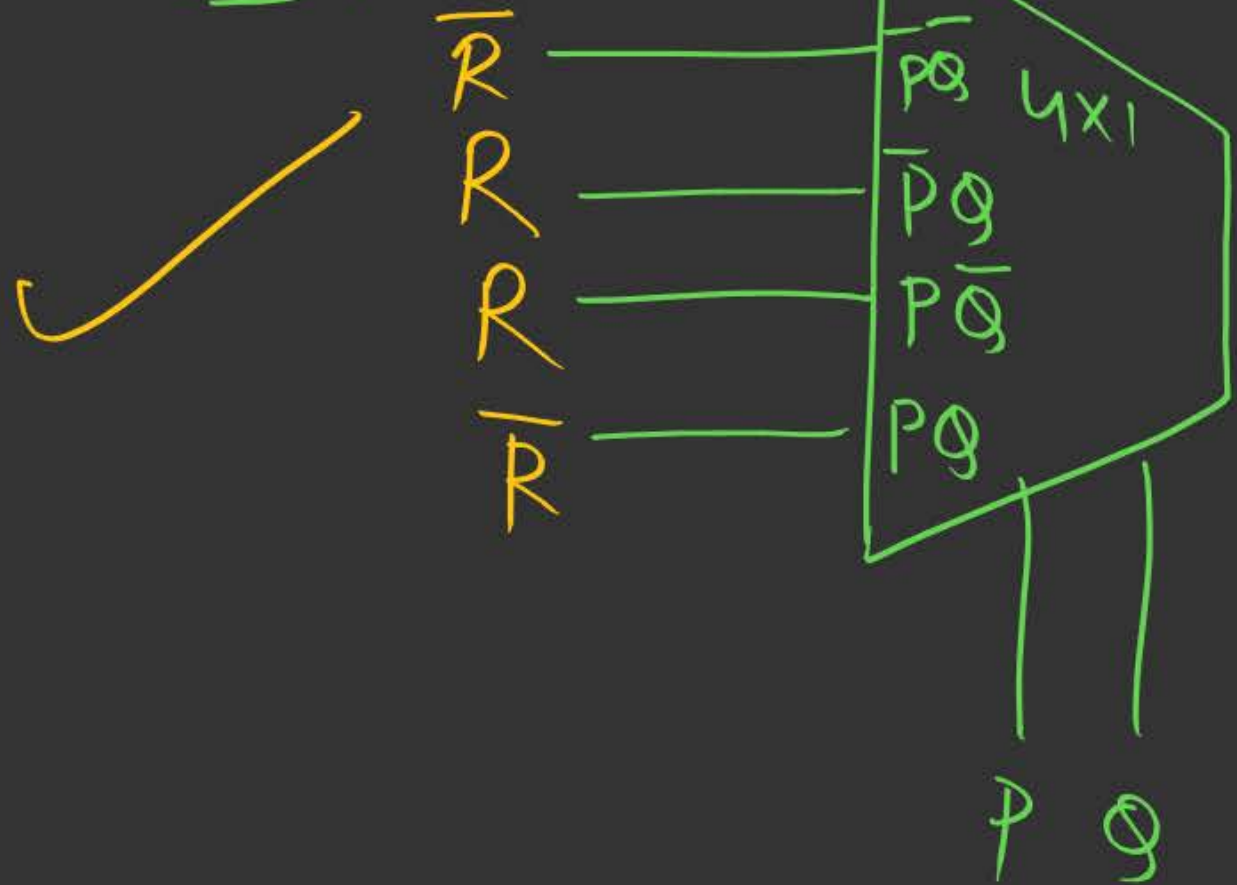
Implementation of a function using MUX.
↓
multiple as per given

(reverse of set 3 process)

eg1): $f(\underline{P, Q, R}) = \overline{P}\overline{Q}\overline{R} + \overline{P}Q\overline{R} + P\overline{Q}\overline{R} + PQR$

Represent / Impl using a 4x1 mux

Soln :-



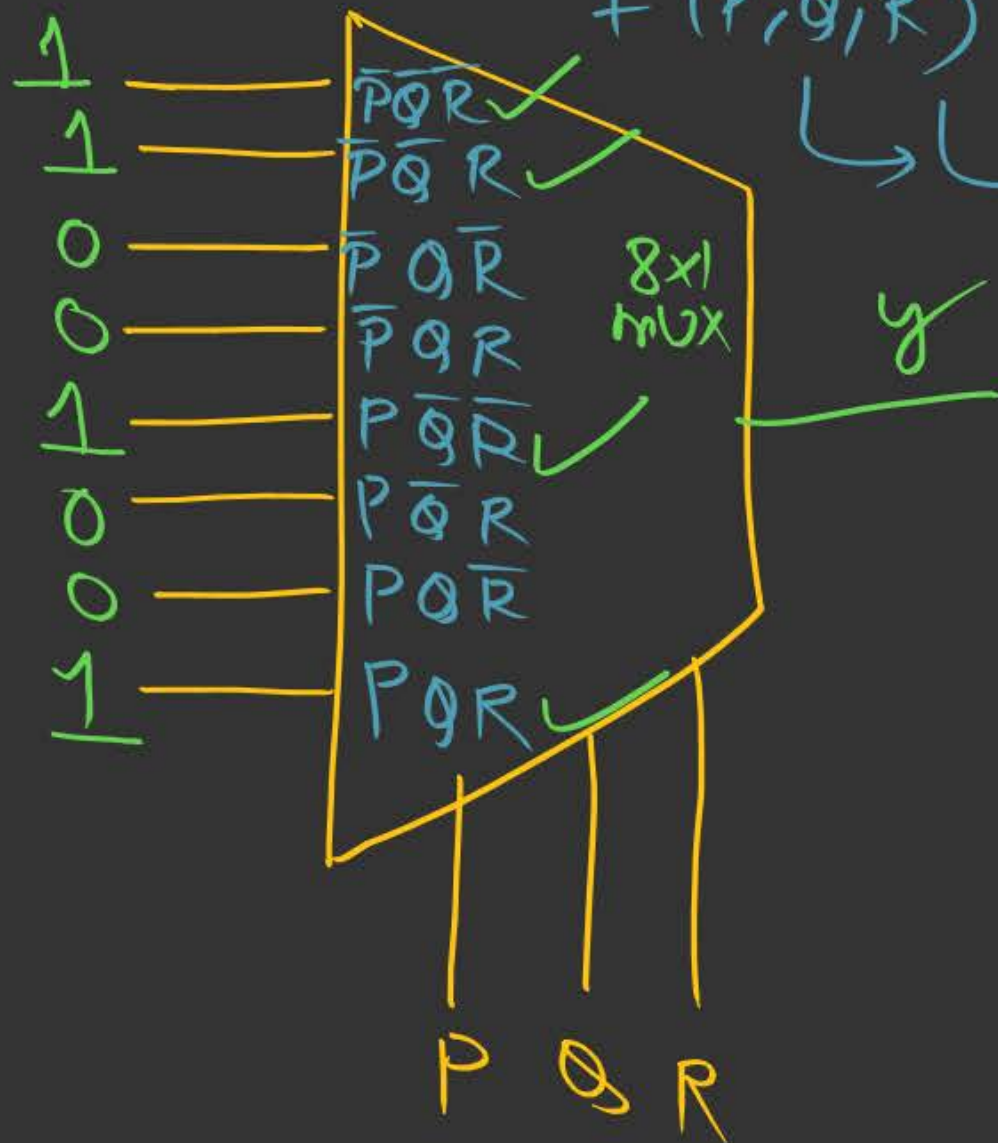
ex2) Represent using 8x1 mux.

$$f(P, Q, R) = \sum m(0, 1, 4, 7)$$

Soln:-

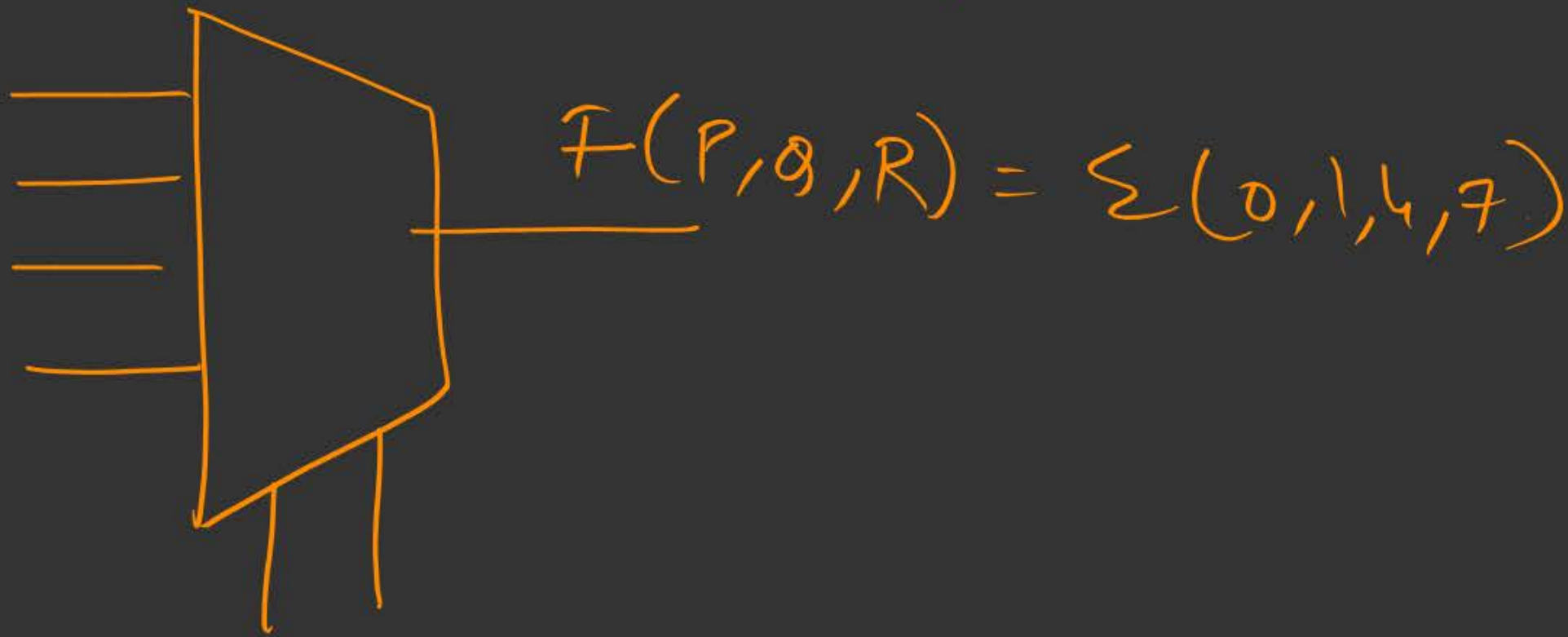
	0	1	4	7
$f(P, Q, R)$	000	001	100	111

$$\rightarrow = \underline{\bar{P}\bar{Q}\bar{R}} + \underline{\bar{P}\bar{Q}R} + \underline{P\bar{Q}\bar{R}} + \underline{PQR}$$



Imp: Design prev case using 4x1 mux.

$$F(P, Q, R) = \bar{P}\bar{Q}\bar{R} + \bar{P}\bar{Q}R + P\bar{Q}\bar{R} + PQR \\ = \Sigma(0, 1, 4, 7)$$



$$\Sigma(0,1,4,7)$$

Case 1:-

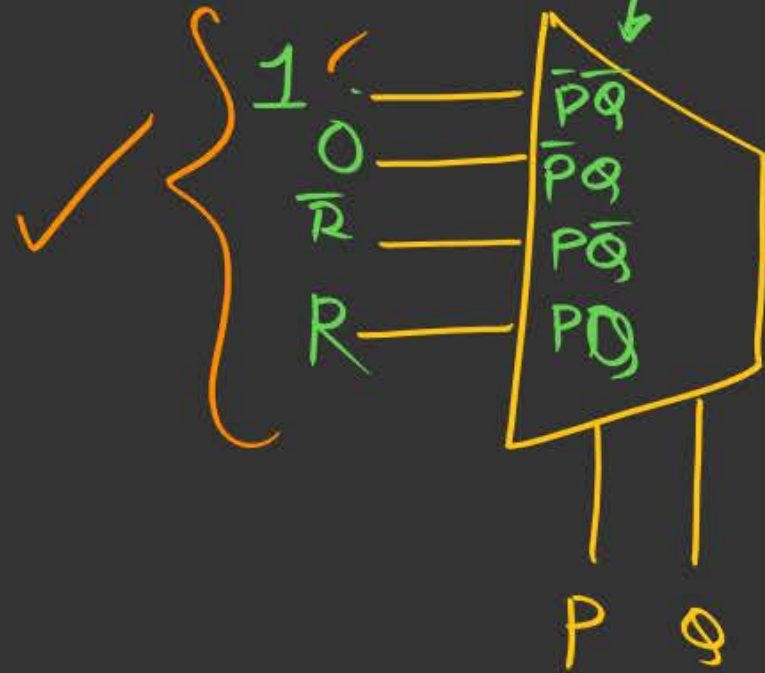
Have PQ as select lines

$$F(P, Q, R) = \Sigma(0, 1, 4, 7)$$

000 001 100 111

$\bar{P}\bar{Q}\bar{R}$ $\bar{P}\bar{Q}R$ $P\bar{Q}\bar{R}$ PQR

Standard



$F(P, Q, R)$

	$\bar{P}\bar{Q}$	$\bar{P}Q$	$P\bar{Q}$	PQ
$\bar{R}$	$\bar{P}\bar{Q}\bar{R}$ 0	$\bar{P}Q\bar{R}$ 2	$P\bar{Q}\bar{R}$ 4	$PQ\bar{R}$ 6
R	$P\bar{Q}R$ 1	$\bar{P}QR$ 3	$P\bar{Q}R$ 5	PQR 7
	1	0	$\bar{R}$	R

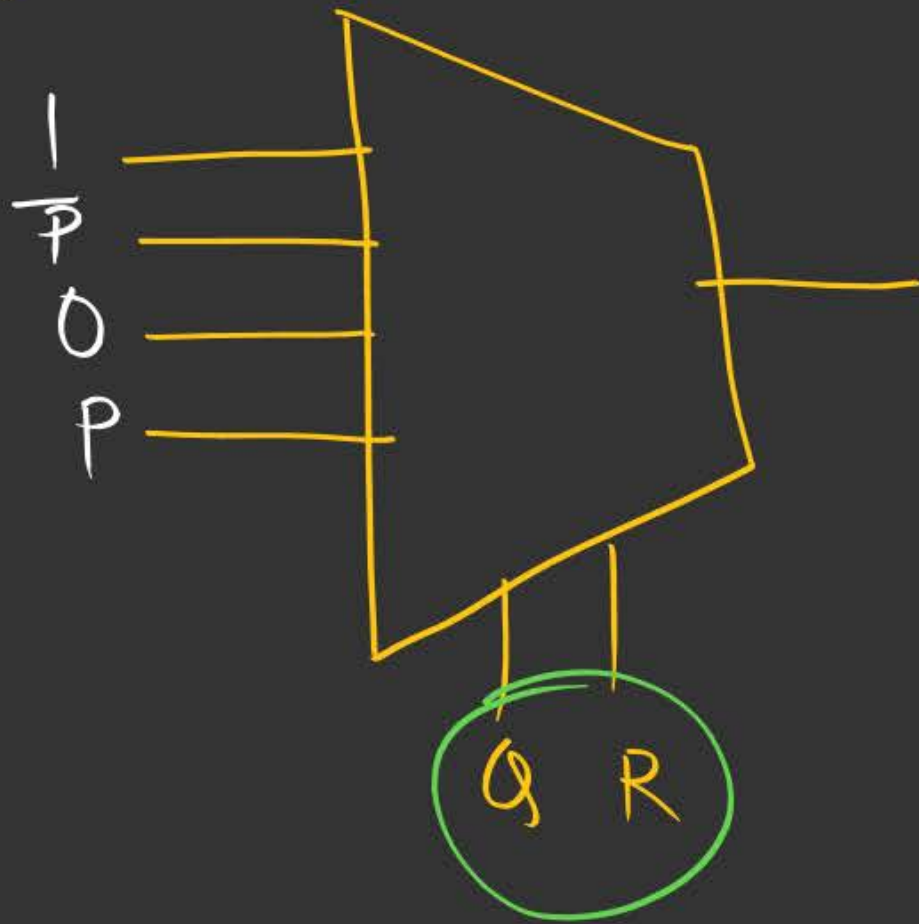
Ques 3:-

$F(P, Q, R)$ using 4×1 mux

same Ques: $\Sigma(0, 1, 4, 7)$

But QR as select lines

Standard:-



	$\overline{Q}\overline{R}$	$\overline{Q}R$	$Q\overline{R}$	QR
$\overline{P}$	000 $\overline{P}\overline{Q}\overline{R}$ 0	001 $\overline{P}\overline{Q}R$ 1	010 $\overline{P}Q\overline{R}$ 2	011 $\overline{P}QR$ 3
P	100 $P\overline{Q}\overline{R}$ 4	101 $P\overline{Q}R$ 5	110 $PQ\overline{R}$ 6	111 PQR 7
	1	$\overline{P}$	0	P

8404cut

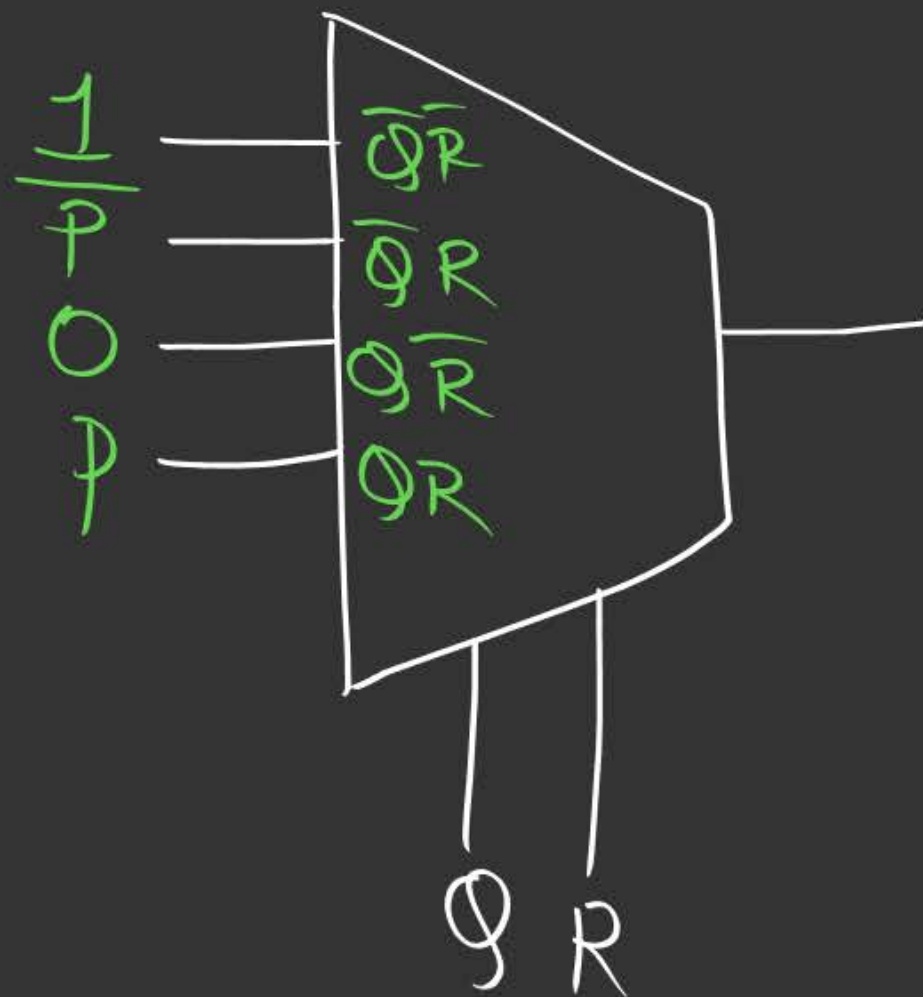
$F(P, Q, R)$

$\Rightarrow \Sigma(0, 1, 4, 7)$

$$= \begin{matrix} 000 & 001 & 100 & 111 \\ \underline{\overline{P}\overline{Q}\overline{R}} & \underline{\overline{P}\overline{Q}R} & \underline{P\overline{Q}\overline{R}} & PQR \end{matrix}$$

$$= \overline{Q}\overline{R}(P + \overline{P}) + \underline{\hspace{2cm}}$$

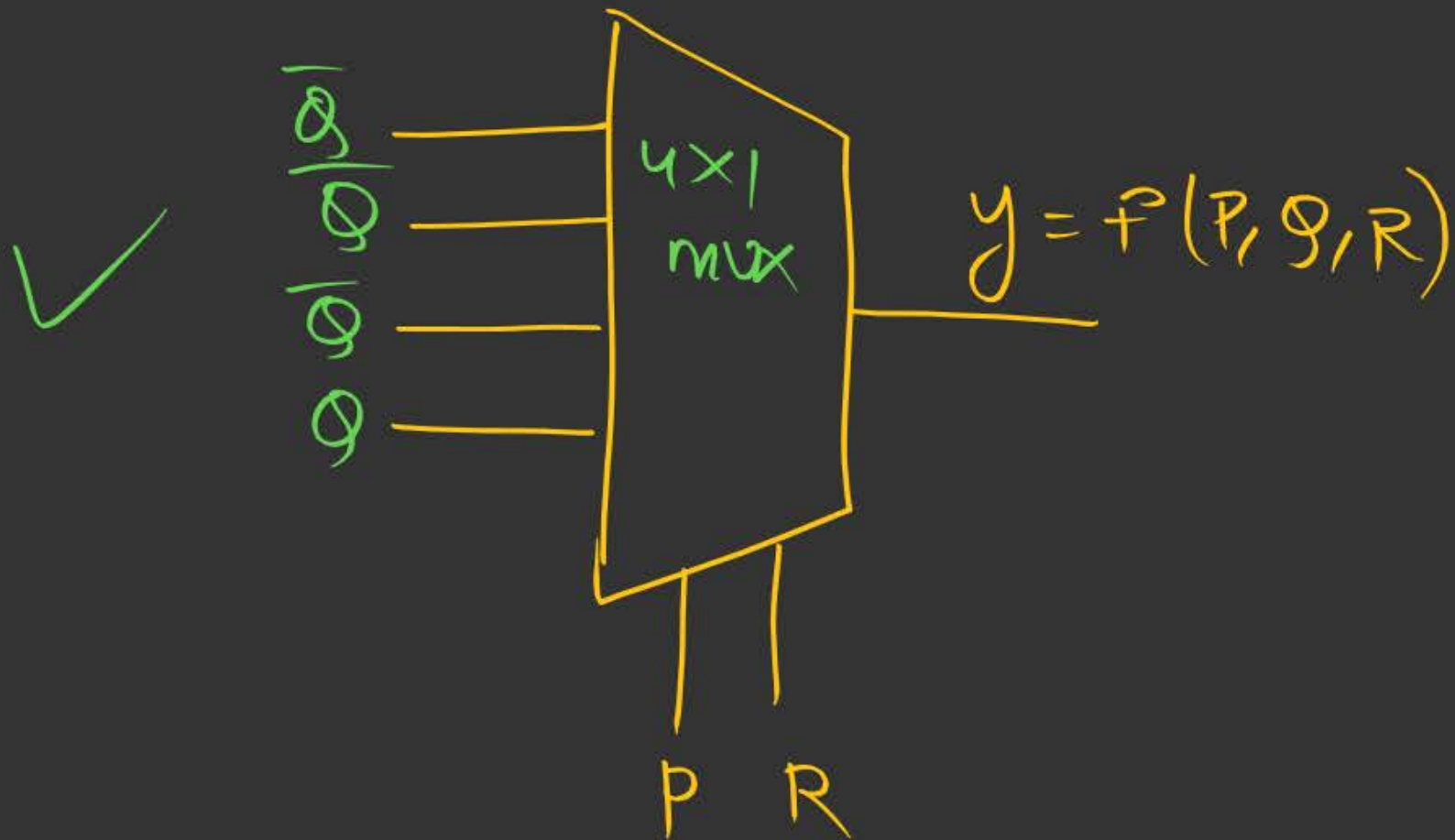
$$= \overline{Q}\overline{R} \cdot 1 + \underline{\hspace{2cm}}$$



$$F(P, Q, R) \subseteq \{0, 1, 4, 7\}$$

Case 4:- Same Ques
But PR as select lines.

Standard:-



(NOT a Kmap)

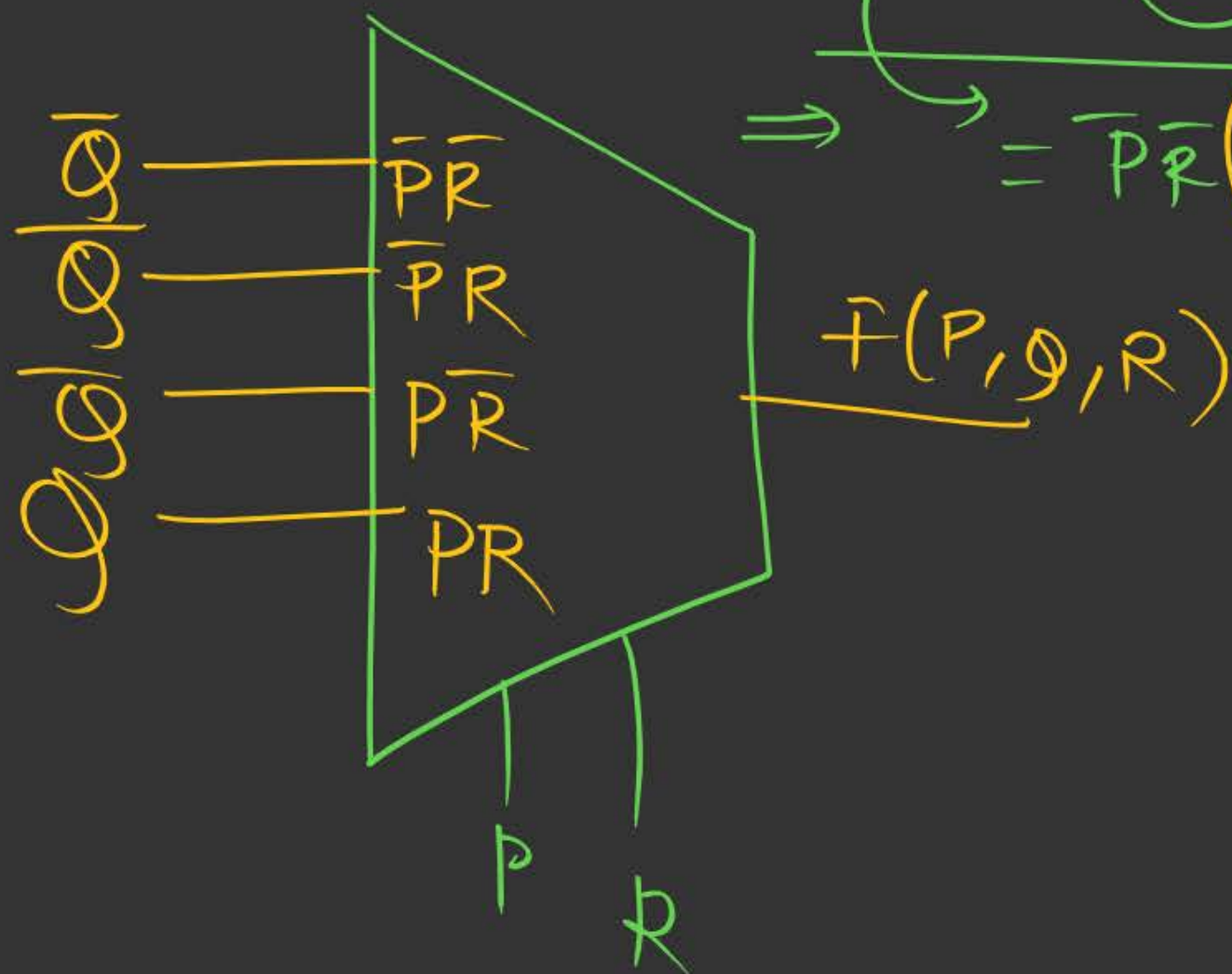
	$\bar{P}\bar{R}$	$\bar{P}R$	$P\bar{R}$	PR
$\bar{Q}$	$\begin{matrix} 000 \\ \bar{P}\bar{Q}\bar{R} \\ 0 \end{matrix}$	$\begin{matrix} 001 \\ \bar{P}\bar{Q}R \\ 1 \end{matrix}$	$\begin{matrix} 100 \\ P\bar{Q}\bar{R} \\ 4 \end{matrix}$	$\begin{matrix} 101 \\ P\bar{Q}R \\ 5 \end{matrix}$
Q	$\begin{matrix} 010 \\ P\bar{Q}\bar{R} \\ 2 \end{matrix}$	$\begin{matrix} 011 \\ P\bar{Q}R \\ 3 \end{matrix}$	$\begin{matrix} 110 \\ PQ\bar{R} \\ 6 \end{matrix}$	$\begin{matrix} 111 \\ PQR \\ 7 \end{matrix}$
	$\bar{Q}$	$\bar{Q}$	$\bar{Q}$	Q

✓ Shortcut:-

$$F(P, Q, R) = \sum (0, 1, 4, 7)$$

$$= \underbrace{\bar{P}\bar{Q}\bar{R}} + \underbrace{\bar{P}\bar{Q}R} + \underbrace{P\bar{Q}\bar{R}} + \underbrace{PQR}$$

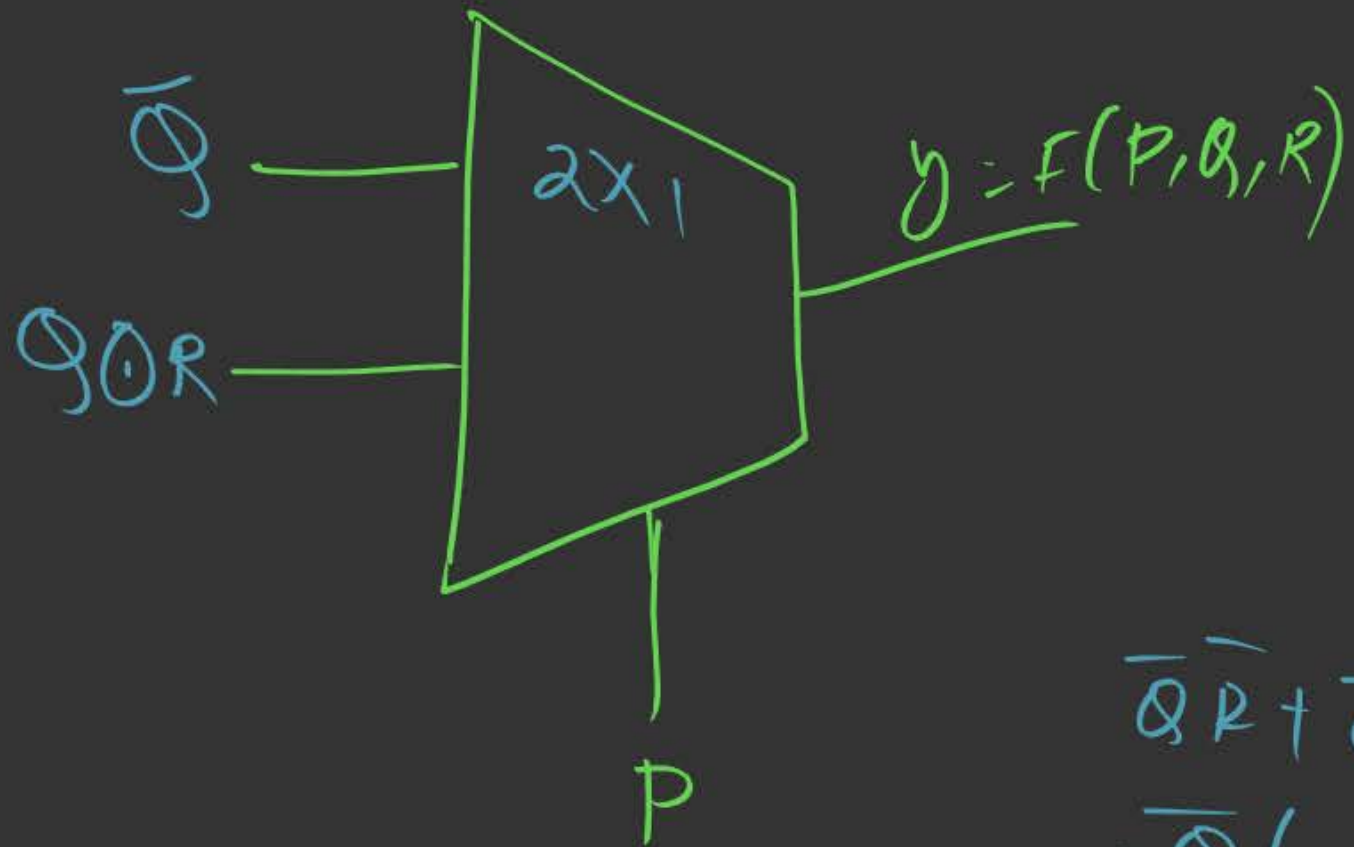
$$\Rightarrow = \bar{P}\bar{R}(\bar{Q}) + \bar{P}R(\bar{Q}) + P\bar{R}(\bar{Q}) + PR(Q)$$



Ques 5) Design using 2x1 mux
with P as select line.

$$F(P, Q, R) = \Sigma(0, 1, 4, 7)$$

Standard



$$\begin{aligned} & \bar{Q}R + \bar{Q}\bar{R} \\ &= \bar{Q}(R + \bar{R}) = \bar{Q} \end{aligned}$$

	$\bar{P}$	P
$\bar{Q}\bar{R}$	$\bar{P}\bar{Q}\bar{R}$ 0 0 0 0	$P\bar{Q}\bar{R}$ 1 0 0 4
$\bar{Q}R$	$\bar{P}\bar{Q}R$ 0 0 1 1	$P\bar{Q}R$ 1 0 1 5
$Q\bar{R}$	$\bar{P}Q\bar{R}$ 0 1 0 2	$PQ\bar{R}$ 1 1 0 6
QR	$\bar{P}QR$ 0 1 1 3	PQR 1 1 1 7
	$\bar{Q}\bar{R} + \bar{Q}R$ $\bar{Q}$	$\bar{Q}\bar{R} + QR$ Q
		$Q \oplus R$

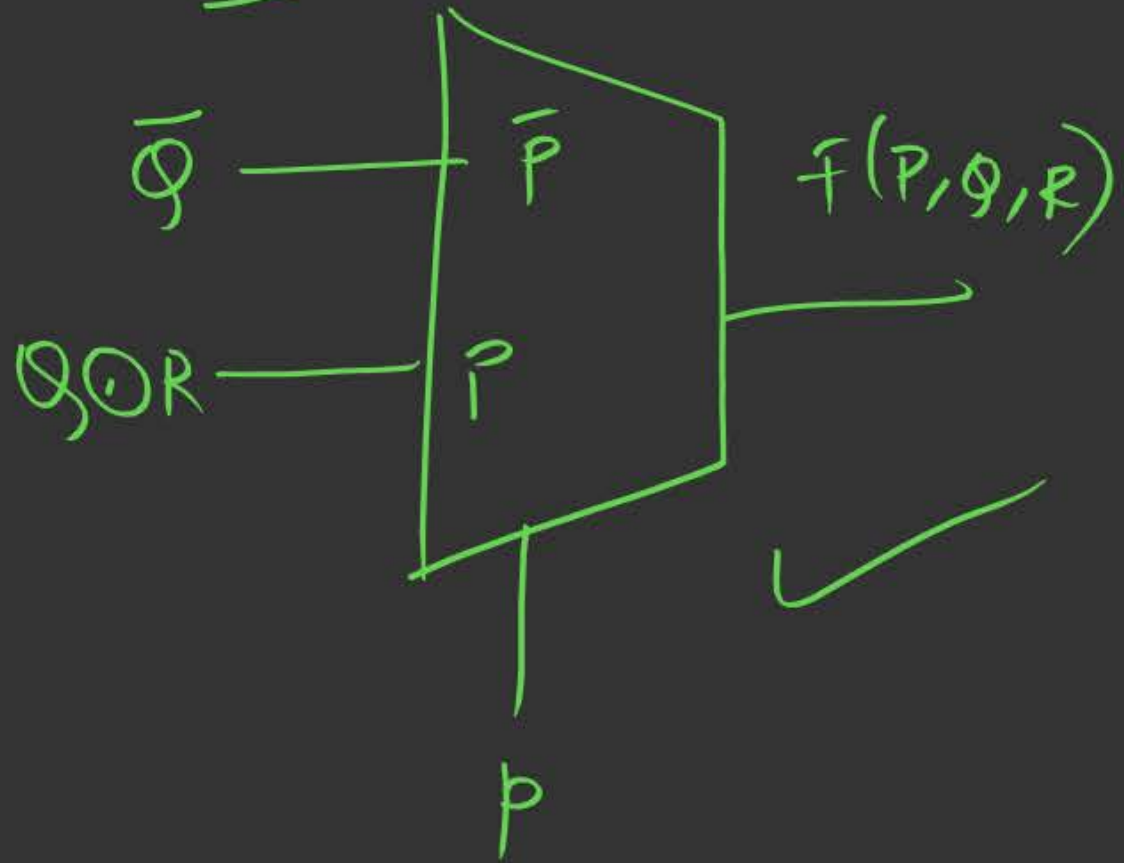
Shortcut

$$F(P, Q, R) \subseteq (0, 1, 4, 7)$$

↓

$$= \overline{P}\overline{Q}\overline{R} + \overline{P}\overline{Q}R + P\overline{Q}\overline{R} + PQR$$

Soln:



$$= \overline{P}(\overline{Q}\overline{R} + \overline{Q}R) + P(\overline{Q}\overline{R} + QR)$$

$$= \overline{P}(\overline{Q}(\overline{R} + R)) + P(Q\overline{R} + QR)$$

$$= \overline{P}\overline{Q} + P(Q\overline{R} + QR)$$



2 mins Summary



Set 3 ✓

Set 4 ✓

Set 5 ✓

Set 6 → 15 miles

Sunday (17th May)

↳ WT-2

↓
[k-maps
& before]



THANK - YOU